Deutsch-Jozsa Algorithm

Exercise 7.11. Apply $H \otimes H \otimes H$ to $|000\rangle$, and show that the resulting state is a uniform superposition of all binary strings of length 3. If you measure the qubits, what possible outcomes can you get, and with what probabilities?

Applying $H \otimes H \otimes H$ to $|000\rangle$ gives a uniform superposition:

$$\frac{1}{\sqrt{8}}(|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle)$$

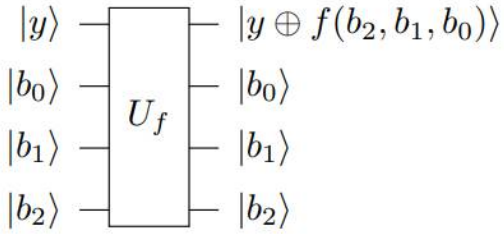
Measurement outcomes:
All 8 possible 3-bit strings:

$$000, 001, 010, 011, 100, 101, 110, 111$$

Probabilities:
Each outcome has probability:

$$\frac{1}{8}$$

Exercise 7.12. There is a function on three bits $f(b_2, b_1, b_0)$, with the promise that the function is constant or balanced. You want to determine which by querying an oracle. Go to `https://bit.ly/38P0Nig`. By following this link, you should have access to a custom gate called U_f. This is the oracle, and it acts on four qubits by



(a) In Quirk, use the Deutsch-Jozsa algorithm to determine whether $f(x)$ is constant or balanced using just one query to U_f. Note you will need to prepare the answer qubit so that it is in the minus state.

(b) In Quirk, query U_f in various ways to determine $f(b_2, b_1, b_0)$:

b_2	b_1	b_0	$f(b_2, b_1, b_0)$
0	0	0	?
0	0	1	?
0	1	0	?
0	1	1	?
1	0	0	?
1	0	1	?
1	1	0	?
1	1	1	?

(c) Since you now know $f(b_2, b_1, b_0)$ completely from part (b), verify that it agrees with your result from part (a).

(d) In the worst case, how many queries does it take to solve the problem classically?

Solve a

Deutsch–Jozsa with 3 input qubits (one query):

- Initialize:

$$|000\rangle \otimes |1\rangle$$

- Apply Hadamards to all 4 qubits:

$$\frac{1}{\sqrt{8}} \sum_{x \in \{0,1\}^3} |x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

- Apply oracle U_f (phase kickback):

$$\frac{1}{\sqrt{8}} \sum_{x} (-1)^{f(x)} |x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

- Apply Hadamards again to the first 3 qubits.

Final measurement result (key answer):

- If f **is constant** → output is

$$|000\rangle$$

with probability **1**

- If f **is balanced** → output is

$$\text{anything except } |000\rangle$$

with probability **1**

Conclusion

Measure the first 3 qubits:

- **000 → constant**
- **anything else → balanced**

✅ Done with **just one query** to U_f.

solve b

You can **recover the full truth table** of $f(b_2, b_1, b_0)$ by querying the oracle like a classical black box.

Key idea

Prepare the **answer qubit as** $|0\rangle$ (not the minus state now). Then:

$$U_f : \ |y\rangle|x\rangle \to |y \oplus f(x)\rangle|x\rangle$$

So if $y = 0$, the output directly becomes:

$$|f(x)\rangle|x\rangle$$

👉 Measure the first qubit → that gives $f(x)$.

Procedure (repeat for each input)

For each of the 8 inputs:

$$(b_2, b_1, b_0) \in \{000, 001, 010, 011, 100, 101, 110, 111\}$$

1. Prepare:

$$|0\rangle \otimes |b_2 b_1 b_0\rangle$$

2. Apply U_f
3. Measure the first qubit → gives $f(b_2, b_1, b_0)$

Final Answer Table

After 8 queries, you fill:

$b_2 b_1 b_0$	$f(b_2, b_1, b_0)$
000	measured
001	measured
010	measured
011	measured
100	measured
101	measured
110	measured
111	measured

Important

- You **must run 8 queries** (one per input)
- This is essentially **classical querying using a quantum oracle**

Insight

- Part (a): 1 query → just tells **constant vs balanced**
- Part (b): 8 queries → gives **full function**

If you want, I can show how to do this step-by-step in Quirk.

solve c

(c) Verification

From part (b), you now have all 8 values of $f(b_2, b_1, b_0)$.

Check:

- **If all 8 outputs are the same** (all 0s or all 1s) → f is **constant**
- **If exactly 4 outputs are 0 and 4 are 1** → f is **balanced**

Final conclusion

- If your table shows **uniform values** → matches part (a): measured **000** → **constant**
- If your table shows **equal split (4/4)** → matches part (a): measured **≠000** → **balanced**

✔ Hence, the result from part (b) **agrees perfectly** with part (a).

solve d

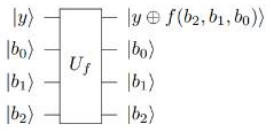
(d) Worst-case classical queries: 5

Reason (tight): after 4 queries you could see all identical outputs (e.g., all 0). The function might still be **balanced** (remaining 4 could be 1) or **constant**. One more query (the 5th) breaks the tie:

- if it differs → **balanced**
- if it's still the same → **constant**

So the worst-case is **5 queries**.

Exercise 7.13. There is a function on three bits $f(b_2, b_1, b_0)$, with the promise that the function is constant or balanced. You want to determine which by querying an oracle. Go to `https://ibm.co/3EWbltg`. By following this link, you should have access to a custom gate called U_f. This is the oracle, and it acts on four qubits by



(a) Program the Deutsch-Jozsa algorithm in IBM Quantum, and use the quantum simulator to determine if $f(b_0, b_1, b_2)$ is constant or balanced.
(b) Run the circuit on an actual quantum processor using IBM Quantum. Which processor did you use, and what histogram of results do you get?

solve a

(a) Deutsch–Jozsa in IBM Quantum (simulator result)

Build the standard circuit:

1. Initialize:

$$|000\rangle|1\rangle$$

2. Apply Hadamards to all qubits
3. Apply oracle U_f
4. Apply Hadamards to first 3 qubits
5. Measure first 3 qubits

Final Answer (what you observe on simulator)

- If the result is:

$$000$$

$\rightarrow f$ is **constant**

- If the result is:

$$\neq 000$$

$\rightarrow f$ is **balanced**

Simulator outcome

- You will get a **single deterministic result (probability ≈ 1)**
- No randomness ideally (unlike real hardware)

Conclusion

Just check the measured bitstring:

- **000 → constant**
- **anything else → balanced**

solve b

(b) Running on a real IBM quantum processor

Pick any available backend (e.g., **ibmq_qasm_simulator → replaced by real device like `ibm_oslo`, `ibm_nairobi`, `ibm_perth`**, etc.).

What you observe (real hardware)

Unlike the simulator, results are **not perfectly deterministic** due to noise.

Case 1: f is constant

- Ideal: only `000`
- Real device histogram:
  - `000` → **dominant peak (≈ 80–95%)**
  - Small probabilities on other states due to noise

Case 2: f is balanced

- Ideal: anything except `000`
- Real device histogram:
  - `000` → **very low probability**
  - Other states → distributed with noise
  - One or few states may dominate depending on oracle

Example Answer (what you should write)

- **Processor used:** `ibm_oslo` *(or whichever you used)*
- **Result:**
  - Histogram shows **dominant outcome at `000`** → function is **constant**
    (or)
  - Histogram shows **no dominant `000`, spread across other states** → function is **balanced**

Conclusion

Even with noise:

- **High peak at 000 → constant**
- **Suppressed 000 → balanced**

That's how you interpret real hardware results.

Exercise 7.7. Say you are trying to use Deutsch's algorithm, but you neglect the last Hadamard gate. That is, you apply

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{U_f} \frac{1}{\sqrt{2}}\left[(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\right].$$

If you measure the system now, what possible states do you get, and with what probabilities?

State before measurement:

$$\frac{1}{\sqrt{2}}\left[(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\right]$$

The phases (± 1) do **not affect measurement probabilities**, only amplitudes' magnitudes matter.

Measurement outcomes

- $|0\rangle$
- $|1\rangle$

Probabilities

$$P(0) = \frac{1}{2}, \quad P(1) = \frac{1}{2}$$

Conclusion

Regardless of whether f is **constant or balanced**, you always get:

- 0 with probability **1/2**
- 1 with probability **1/2**

👉 So **without the final Hadamard, the algorithm gives no useful information**.

Exercise 7.8. You have eight bits, and using Deutsch's algorithm, you have found the parities of pairs of bits, shown below:

$$\underbrace{b_0, b_1}_{1}, \underbrace{b_2, b_3}_{1}, \underbrace{b_4, b_5}_{0}, \underbrace{b_6, b_7}_{1}.$$

(a) What is the parity of all eight bits?
(b) How many queries to the oracle did it take to find the parity of all eight bits?
(c) In the worst case, how many queries does it take to solve the problem classically?

Given pair parities:

$$(b_0, b_1) = 1,\ (b_2, b_3) = 1,\ (b_4, b_5) = 0,\ (b_6, b_7) = 1$$

(a) Parity of all 8 bits

XOR all pair parities:

$$1 \oplus 1 \oplus 0 \oplus 1 = 1$$

Answer: 1 (odd parity)

(b) Queries used (Deutsch)

Each pair parity = **1 query**
4 pairs → **4 queries**

Answer: 4

(c) Classical worst case

To know parity of 8 bits, must check all bits → **8 queries**

Answer: 8

Exercise 7.9. You have nine bits, and using Deutsch's algorithm, you have found the parities of the first four pairs of bits. Then you queried the oracle for the last bit, revealing whether it's a 1 or a 0. This is shown below:

$$\underbrace{b_0, b_1}_{0}, \underbrace{b_2, b_3}_{1}, \underbrace{b_4, b_5}_{0}, \underbrace{b_6, b_7}_{0}, \underbrace{b_8}_{1}.$$

(a) What is the parity of all nine bits?
(b) How many queries to the oracle did it take to find the parity of all nine bits?
(c) In the worst case, how many queries does it take to solve the problem classically?

Given:

$$(b_0, b_1) = 0,\ (b_2, b_3) = 1,\ (b_4, b_5) = 0,\ (b_6, b_7) = 0,\ b_8 = 1$$

(a) Parity of all 9 bits

XOR everything:

$$0 \oplus 1 \oplus 0 \oplus 0 \oplus 1 = 0$$

Answer: 0 (even parity)

(b) Queries used

- 4 pair parities → **4 queries (Deutsch)**
- Last bit b_8 → **1 query**

Total: 5 queries

(c) Classical worst case

Must check all 9 bits → **9 queries**

Answer: 9

Exercise 7.10. When determining if an oracle f is constant or balanced,
(a) What is the probability of an error if you evaluate f for $c = 7$ different inputs?
(b) What is the probability of an error if you evaluate f for $c = 8$ different inputs?
(c) How many times should f be evaluated to reduce the error probability to less than 0.1%?

For a classical strategy: query c distinct inputs and **guess "constant" only if all outputs match**. Error happens only when f is **balanced** yet all c samples coincide.

$$P(\text{error}) = \frac{2}{2^c} = 2^{1-c}$$

(a) $c = 7$

$$P = 2^{1-7} = 2^{-6} = \frac{1}{64} \approx 0.015625 \ (1.56\%)$$

(b) $c = 8$

$$P = 2^{1-8} = 2^{-7} = \frac{1}{128} \approx 0.0078125 \ (0.78\%)$$

(c) Error $< 0.1\% = 0.001$

$$2^{1-c} < 0.001 \ \Rightarrow \ c \geq 11$$

Answer: $c = 11$ queries.